

Objective:

Establish a GCP environment aligned with SOC 2 controls, demonstrating secure infrastructure configuration, network monitoring, IAM governance, and audit logging. Using GCP official documentation.

Goals:

- Implement IAM role for administrative and operational access
 - Configure network segmentation and firewall rules for restricted access
 - Enable VPC Flow Logs for monitoring ingress/egress traffic
 - Enable Security Command Center for threat detection and continuous monitoring
 - Collect and query Cloud Logging for audit and compliance verification
 - Validate secure instance configuration and logging visibility
-

Tools & Services Used**Google Cloud Platform Services:**

- Compute Engine
- VPC Networks & Subnets
- Firewall Rules
- Security Command Center
- Cloud Logging / Audit Logs

Linux Environment:

Ubuntu 22.04 LTS

Network & Security Utilities:

`iptables`, `ss`, `ping`, `tcpdump`, `ufw` / `firewal-lrules` (GCP)

Command-line Tools:

`gcloud`, `ssh`, `vim`, `head`, `grep`

GCP Services & Integration

`Cloud Logging`, `IAM`, `VPC Flow Logs`, `Security Command Center`, `Compute Engine`

1. Set Up GCP Project & Compute Instance (Confirming project and instance is up and running)

```
alihammam0178@cloudshell:~ (soc2-1)$ gcloud config get project
Your active configuration is: [cloudshell-10490]
soc2-1
alihammam0178@cloudshell:~ (soc2-1)$ gcloud compute instances list
NAME: soc2lab
ZONE: us-central1-a
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.128.0.2
EXTERNAL_IP: 34.16.127.162
STATUS: RUNNING
```

2. Enable VPC Flow Logs and Verification

```
alihammam0178@cloudshell:~ (soc2-1)$ gcloud compute networks subnets update default \
> --region=us-central1 \
> --enable-flow-logs\
>
Updated [https://www.googleapis.com/compute/v1/projects/soc2-1/regions/us-central1/subnetworks/default].
```

Verification

```
alihammam0178@cloudshell:~ (soc2-1)$ gcloud compute networks subnets describe default \
> --region=us-central1 \
> --format="get(logConfig.enable,logConfig.aggregationInterval,logConfig.flowSampling)"
True      INTERVAL_5_SEC  0.5
```

3. Listing Project Logs and Pulling Example Log Entry

Listing project logs

```
alihammam0178@cloudshell:~ (soc2-1)$ gcloud logging logs list --project=soc2-1
NAME: projects/soc2-1/logs/GCEGuestAgent
NAME: projects/soc2-1/logs/OSConfigAgent
NAME: projects/soc2-1/logs/cloudaudit.googleapis.com%2Factivity
NAME: projects/soc2-1/logs/cloudaudit.googleapis.com%2Fsystem_event
NAME: projects/soc2-1/logs/compute.googleapis.com%2Fshielded_vm_integrity
NAME: projects/soc2-1/logs/compute.googleapis.com%2Fvpc_flows
NAME: projects/soc2-1/logs/diagnostic-log
NAME: projects/soc2-1/logs/google_metadata_script_runner
```

Log excerpt presented for brevity (20 lines shown).

```
alihammam0178@cloudshell:~ (soc2-1)$ gcloud logging read 'logName="projects/soc2-1/logs/diagnostic-log"' --limit=5 --format=json | head -n 20
[
  {
    "insertId": "ibe2nrf2tamdb",
    "jsonPayload": {
      "logging.googleapis.com/diagnostic": {
        "instrumentation_source": {
          {
            "name": "go",
            "version": "1.7.0"
          }
        },
        "runtime": "1.21.1"
      }
    },
    "labels": {
      "instance_name": "soc2lab"
    },
    "logName": "projects/soc2-1/logs/diagnostic-log",
    "receiveTimestamp": "2025-11-05T22:57:38.870708052Z",
    "resource": {
```

4. Firewall setup (least privilege principle, only mandatory ports open)

```
alihammam0178@cloudshell:~ (soc2-1)$ gcloud compute firewall-rules list
NAME: default-allow-icmp
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 65534
ALLOW: icmp
DENY:
DISABLED: False

NAME: default-allow-internal
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 65534
ALLOW: tcp:0-65535,udp:0-65535,icmp
DENY:
DISABLED: False

NAME: default-allow-rdp
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 65534
ALLOW: tcp:3389
DENY:
DISABLED: False

NAME: default-allow-ssh
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 65534
ALLOW: tcp:22
DENY:
DISABLED: False
```

5. IAM Role Creation

Created example admin role with permission

```
alihammam0178@cloudshell:~ (soc2-1)$ gcloud iam roles create SOC2_Lab_Admin \
> --project=soc2-1 \
> --title="SOC2 Ali Admin" \
> --permissions=compute.instances.get,compute.instances.list,compute.firewalls.list,compute.networks.list,compute.subnetwork
s.list,logging.logEntries.list,securitycenter.findings.list \
> --stage=GA

Created role [SOC2_Lab_Admin].
etag: BwZC8RPOTVQ=
includedPermissions:
- compute.firewalls.list
- compute.instances.get
- compute.instances.list
- compute.networks.list
- compute.subnetworks.list
- logging.logEntries.list
- securitycenter.findings.list
name: projects/soc2-1/roles/SOC2_Lab_Admin
stage: GA
title: SOC2 Ali Admin
```

```
description: Example Admin role for SOC2 Compliance
etag: BwZC8PS7xEo=
name: projects/soc2-1/roles/soc2AliAdmin
title: SOC2 Ali Admin
```

Assigning/Binding role to my user

```
alihammam0178@cloudshell:~ (soc2-1)$ gcloud projects get-iam-policy soc2-1 \
> --flatten="bindings[].members" \
> --filter="bindings.members:alihammam0178"
---
bindings:
  members: user:AliHammam0178@gmail.com
  role: projects/soc2-1/roles/SOC2_Lab_Role
etag: BwZC4nVv4do=
version: 1
---
bindings:
  members: user:AliHammam0178@gmail.com
  role: projects/soc2-1/roles/soc2AliAdmin
etag: BwZC4nVv4do=
version: 1
---
bindings:
  members: user:AliHammam0178@gmail.com
  role: roles/owner
etag: BwZC4nVv4do=
version: 1
```

6. Enabling Security Command Center

```
alihammam0178@cloudshell:~ (soc2-1)$ gcloud services enable securitycenter.googleapis.com --project=soc2-1
alihammam0178@cloudshell:~ (soc2-1)$
```

Confirming Security API's

```
alihammam0178@cloudshell:~ (soc2-1)$ gcloud services list --enabled | grep security
NAME: securitycenter.googleapis.com
NAME: securitycentermanagement.googleapis.com
```

7. Showing enabled services

```
alihammam0178@cloudshell:~ (soc2-1)$ gcloud services list --enabled
NAME: analyticshub.googleapis.com
TITLE: Analytics Hub API

NAME: bigquery.googleapis.com
TITLE: BigQuery API

NAME: bigqueryconnection.googleapis.com
TITLE: BigQuery Connection API

NAME: bigquerydatapolicy.googleapis.com
TITLE: BigQuery Data Policy API

NAME: bigquerydatatransfer.googleapis.com
TITLE: BigQuery Data Transfer API

NAME: bigquerymigration.googleapis.com
TITLE: BigQuery Migration API

NAME: bigqueryreservation.googleapis.com
TITLE: BigQuery Reservation API

NAME: bigquerystorage.googleapis.com
TITLE: BigQuery Storage API

NAME: cloudapis.googleapis.com
TITLE: Google Cloud APIs

NAME: cloudtrace.googleapis.com
TITLE: Cloud Trace API

NAME: compute.googleapis.com
TITLE: Compute Engine API

NAME: dataform.googleapis.com
TITLE: Dataform API

NAME: dataplex.googleapis.com
TITLE: Cloud Dataplex API

NAME: datastore.googleapis.com
TITLE: Cloud Datastore API

NAME: logging.googleapis.com
TITLE: Cloud Logging API

NAME: monitoring.googleapis.com
TITLE: Cloud Monitoring API

NAME: oslogin.googleapis.com
TITLE: Cloud OS Login API

NAME: securitycenter.googleapis.com
TITLE: Security Command Center API

NAME: securitycentermanagement.googleapis.com
TITLE: Security Command Center Management API

NAME: servicemanagement.googleapis.com
TITLE: Service Management API

NAME: serviceusage.googleapis.com
TITLE: Service Usage API

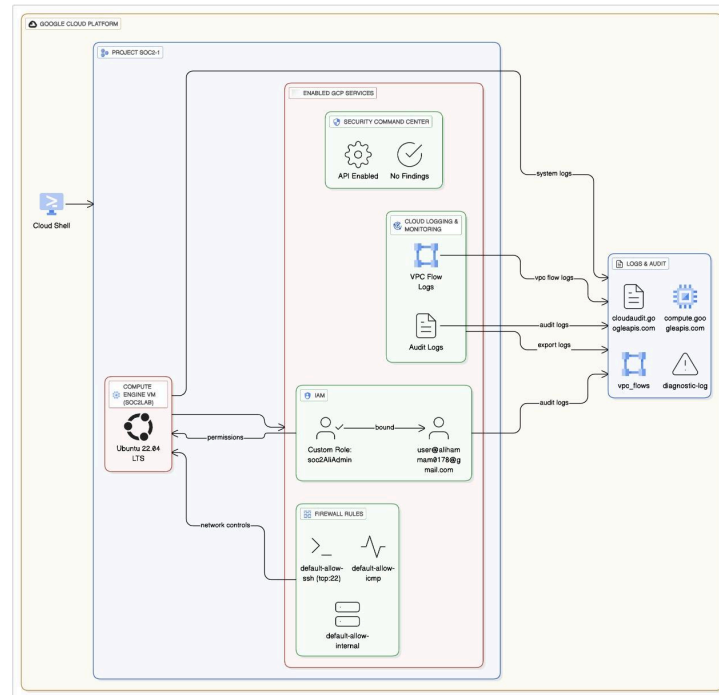
NAME: sql-component.googleapis.com
TITLE: Cloud SQL

NAME: storage-api.googleapis.com
TITLE: Google Cloud Storage JSON API

NAME: storage-component.googleapis.com
TITLE: Cloud Storage

NAME: storage.googleapis.com
TITLE: Cloud Storage API
alihammam0178@cloudshell:~ (soc2-1)$
```

Diagram



Result

- Deployed and configured a single VM (**soc21ab**) in GCP for SOC 2 simulation.
- Enabled Cloud Logging and verified audit and system logs for activity visibility.
- Confirmed VPC Flow Logs were active, capturing network traffic metadata.
- Created and bound custom IAM roles for access control.
- Verified Security Command Center APIs were enabled and accessible.
- Collected evidence of IAM bindings, enabled services, and firewall configurations.
- Least-privilege access enforcement through custom IAM role testing.
- Generated log queries and outputs as compliance verification evidence.